Using clinical features to predict likelihood of a stroke in patients

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Abstract

This project investigates whether a patient's likelihood of stroke can be predicted using a set of clinical features from the Stroke Prediction Dataset. The dataset contains 5,110 observations and includes variables such as age, gender, hypertension, heart disease, residence type, average glucose level, BMI, smoking status, work type, marital status, and prior stroke occurrence. After cleaning and wrangling the data, we conducted exploratory data analysis to identify patterns between these predictors and stroke outcomes, then developed a k-nearest neighbors classification model using an 80/20 train-test split and 5-fold cross-validation for model tuning.

Our exploratory analysis suggested that age, hypertension, heart disease, BMI, and average glucose level were the most informative predictors, while variables such as id, work type, and marital status contributed little to model performance. The final model achieved high overall accuracy on the test set, but the dataset was heavily imbalanced and the model classified nearly all cases as "No stroke," failing to meaningfully identify positive stroke cases.

These findings suggest that future work should consider alternative modeling strategies, better methods for handling class imbalance, and additional predictors related to lifestyle, genetics, and medical history.

Introduction

According to the National Cancer Institute, a stroke occurs when brain tissue is damaged by a loss of blood flow to certain parts of the brain (National Cancer Institute, n.d.). It is a significant global health issue, with an annual mortality rate of 5.5 million people worldwide, making it the second leading cause of death. Understanding the factors that contribute to stroke occurrence is crucial for prevention and effective treatment. Age is widely recognized as the strongest determinant of stroke risk, with the likelihood of experiencing a stroke doubling

every decade after the age of 55. Additionally, hypertension has been identified as the leading risk factor of stroke in both developing and developed nations (Donker 2018).

Our research question explores the ability to accurately predict the likelihood of a stroke in a patient given a set of clinical features. Utilizing the Stroke Prediction Dataset, which comprises of 5,110observations. We aim to identify patterns among these features and aid in the development of prevention strategies.

Methods and Results

Our exploratory data analysis exclusively uses the training data set. Summary statistics for categorical and numeric variables are presented in Table 1.

Table 1: Summary statistics for the variables in the stroke training dataset.

column_name	key	value
age	mean	48.48796
bmi	mean	30.30398
avg_glucose_level	mean	109.01402
gender	Female	1685.00000
gender	Male	1055.00000
residence_type	Rural	1331.00000
residence_type	Urban	1409.00000
smoking_status	Formerly	666.00000
smoking_status	Never	1474.00000
smoking_status	Smokes	600.00000
stroke	No	2595.00000
stroke	Yes	145.00000
hypertension	No	2432.00000
hypertension	Yes	308.00000
heart_disease	No	2573.00000
heart_disease	Yes	167.00000
work_type	Government	385.00000
work_type	Never Worked	12.00000
work_type	Private	1798.00000
work_type	Self-employed	491.00000
work_type	children	54.00000
ever_married	No	659.00000
ever_married	Yes	2081.00000

Visualizations Feature Selection

The `id` feature was removed as it has no biological impact on stroke occurrence. We employed `ggpairs` to identify correlations (Figure 1).

Figure 1: Pairwise relationships among selected stroke predictors

Demographic and Clinical Predictors

As seen in Figure 2, there was no significant difference between males and females in terms of the percentage of stroke cases.

With regards to age (Figure 3), the proportion of stroke cases increases noticeably after age 50, suggesting it is one of the strongest predictors.

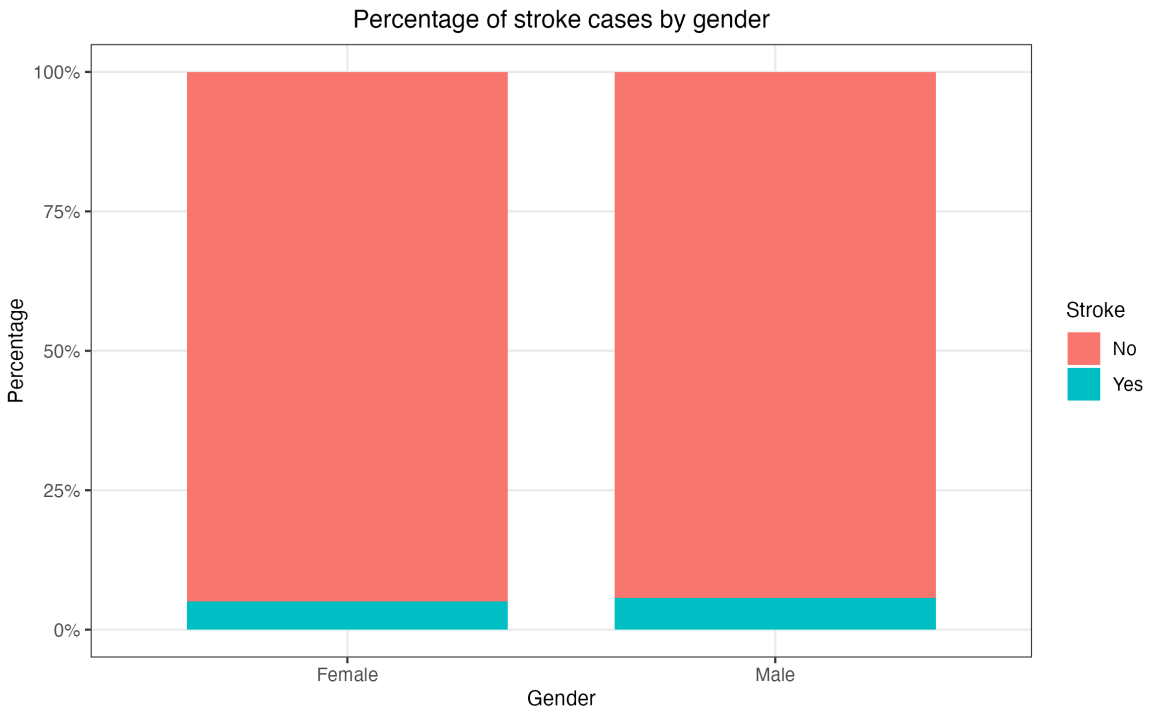

Figure 2: Percentage of stroke cases by gender

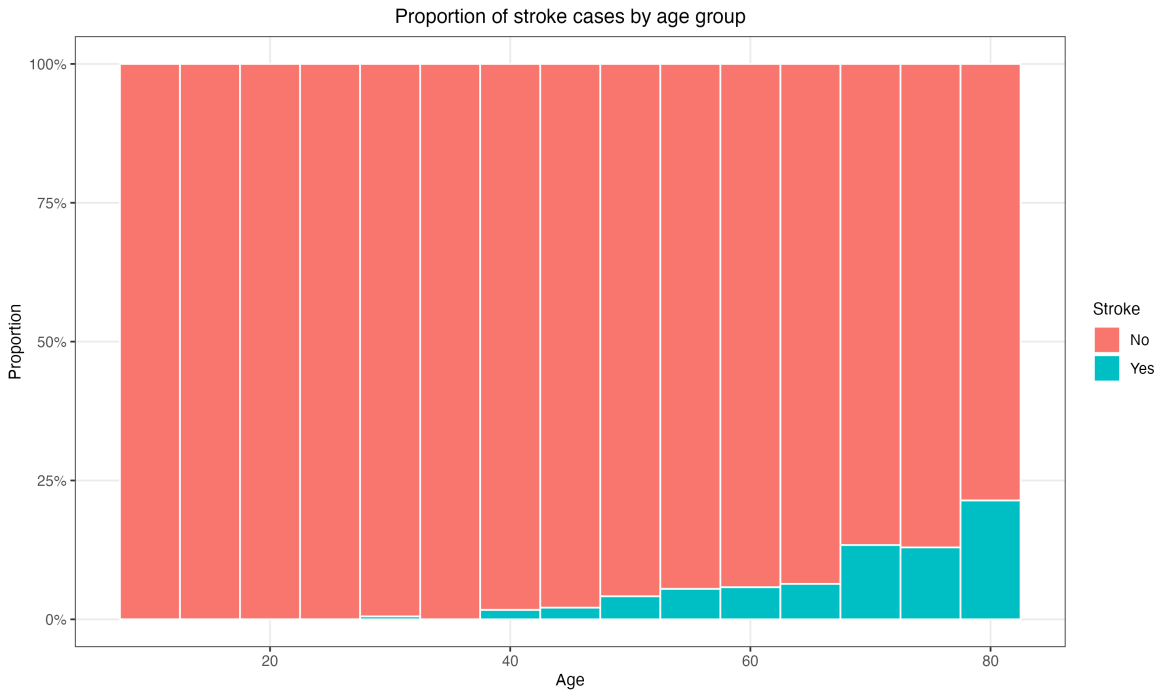

Figure 3: Proportion of stroke cases by age group

Patients with hypertension (Figure 4) and heart disease (Figure 5) are roughly twice as likely to have a stroke.

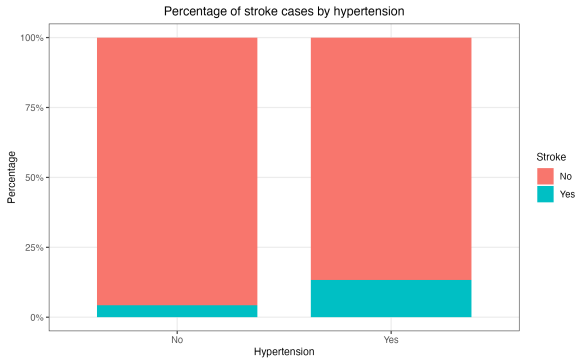

Figure 4: Hypertension

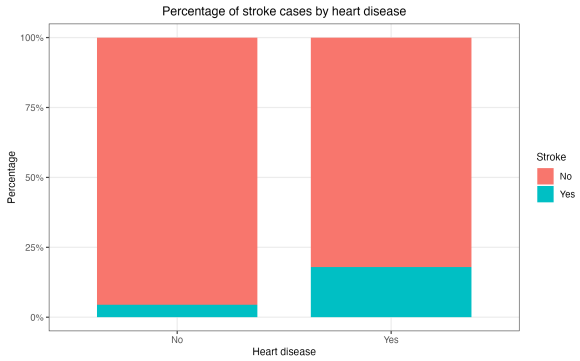

Figure 5: Heart Disease

Conversely, there is no significant difference in likelihood for those residing in urban versus rural areas (Figure 6).

Metabolic Factors

Patients who experienced a stroke tend to have higher average glucose levels (Figure 7) and slightly higher BMI (Figure 8).

Building The Model

Finalized Features

We moved forward with the features defined in our preprocessing script. We excluded `id`, `ever_married`, and `work_type` as they were determined to be too broad or uninformative.

Model Comparison and Tuning

kNN Model

For the **kNN model**, we first performed a coarse sweep of neighbors from 0 to 200 stepping by 10. We found that performance was best at k=171 (Figure 9).

We then performed a finer sweep around the optimal k value, testing neighbors from 155 to 175 stepping by 1. The best performance was achieved at k=165.

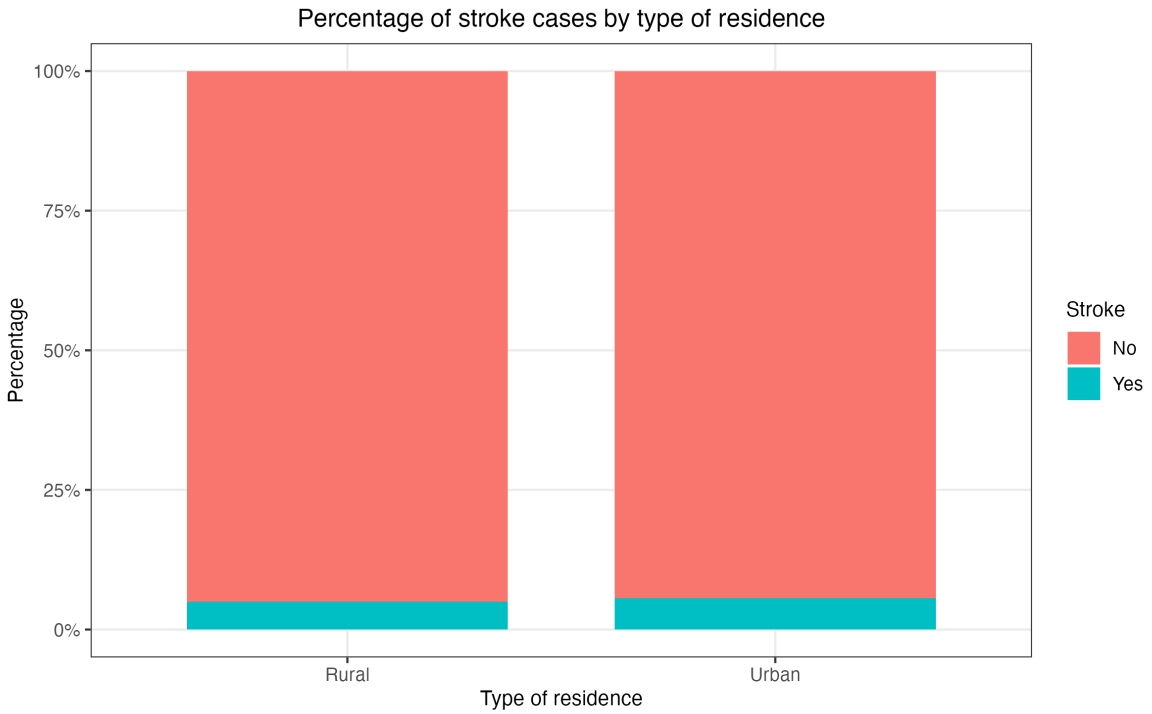

Figure 6: Percentage of stroke cases by type of residence

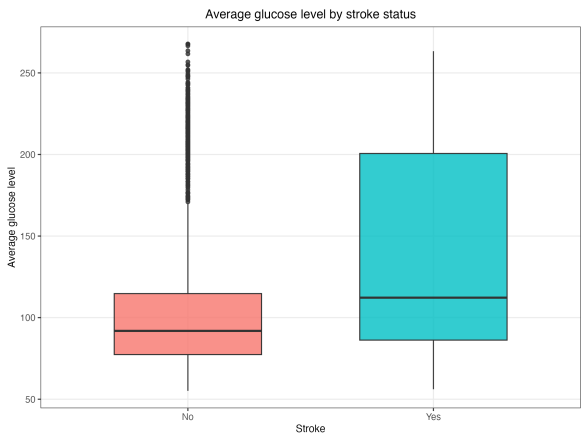

Figure 7: Average glucose level by stroke status

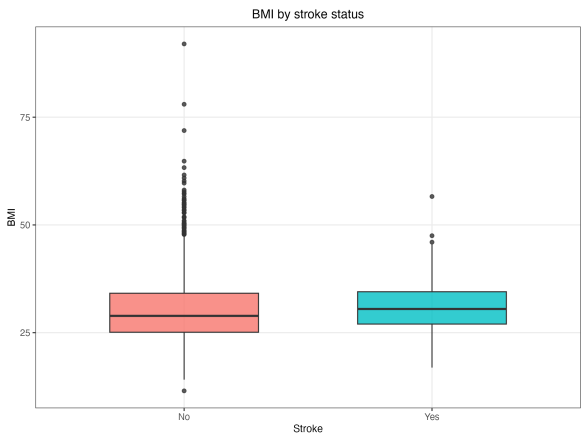

Figure 8: BMI by stroke status

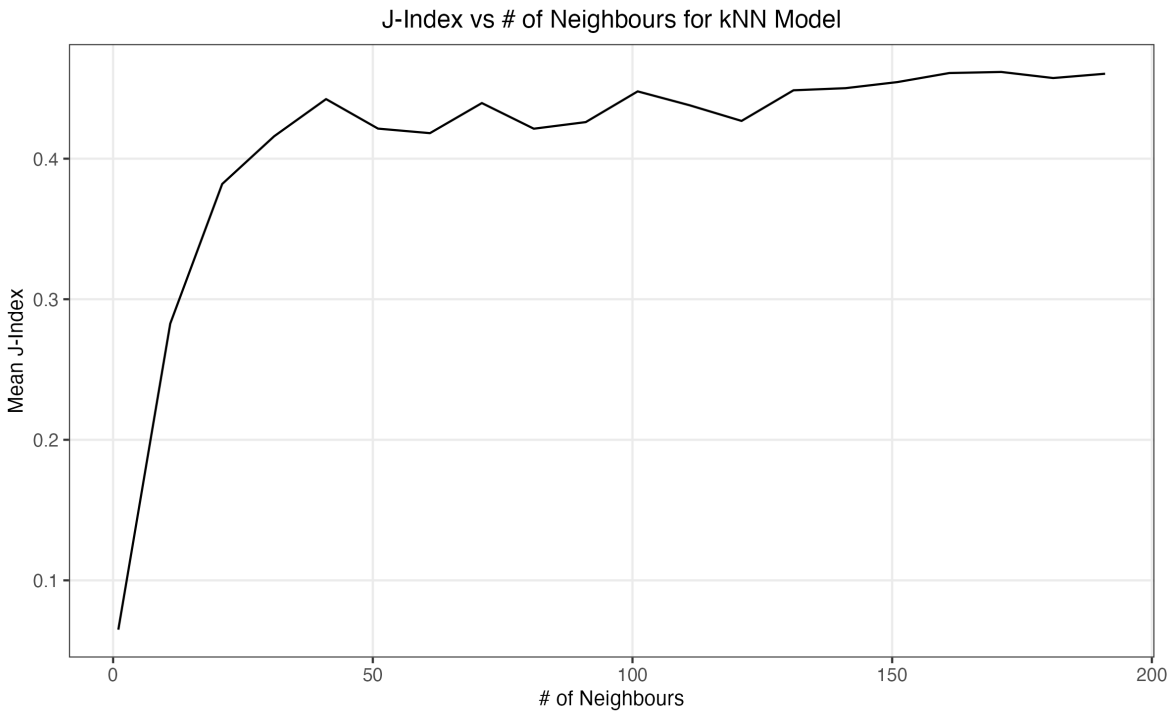

Figure 9: J-Index vs # of Neighbours for kNN Model

Evaluating the final kNN model on the validation set yielded a J-index of 0.2628.

Logistic Regression

For the logistic regression model, we performed backwards selection to identify the most informative predictors. Age, average glucose level, and the presence of pre-existing heart disease emerged as the strongest predictors of stroke. We then tuned the penalty and mixture parameters for the final model, with the best performance achieved at a penalty of 1 and a mixture of 0. Evaluating the final logistic regression model on the validation set yielded a J-index of 0.4714.

XGBoost

For the **XGBoost model**, we analyzed feature importance. Age and average glucose level emerged as the most critical features (Figure 10).

We then perform hyperparameter tuning on several key parameters such as number of trees, tree depth, learning rate, and minimum number of observations in a node. The best performance was achieved with 635 trees, a tree depth of 6, a learning rate of 0.0014, and a minimum number of observations in a node of 38. Evaluating the final XGBoost model on the validation set yielded a J-index of 0.5036.

Model Selection

We compared the three models on the validation set using J-index, sensitivity, and accuracy (Figure 11).

Inspecting the confusion matrices in Figure 12, we can evaluate how each model handled false positives and false negatives.

Based on the highest J-Index, the **XGBoost** model was selected as the best performing architecture.

Final Evaluation

The best model was evaluated on the held-out test set. It achieved a final J-index of **0.5072**.

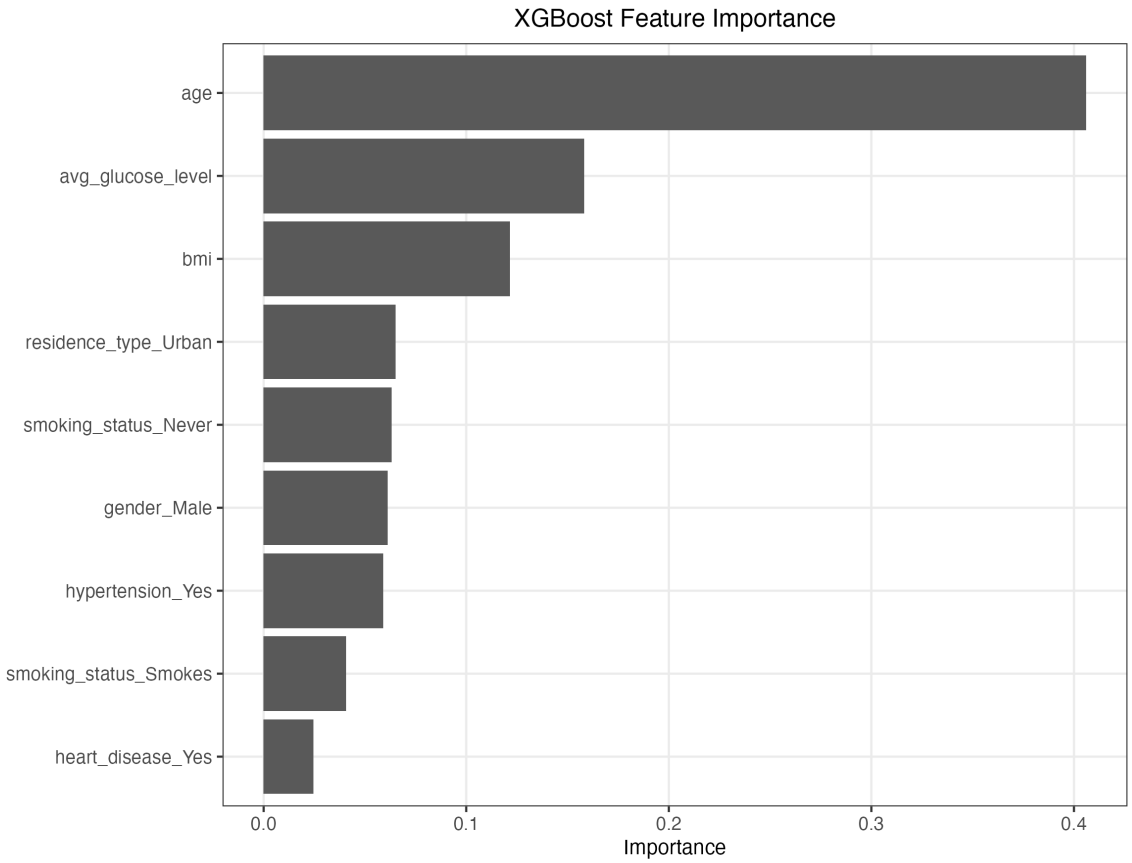

Figure 10: XGBoost Feature Importance

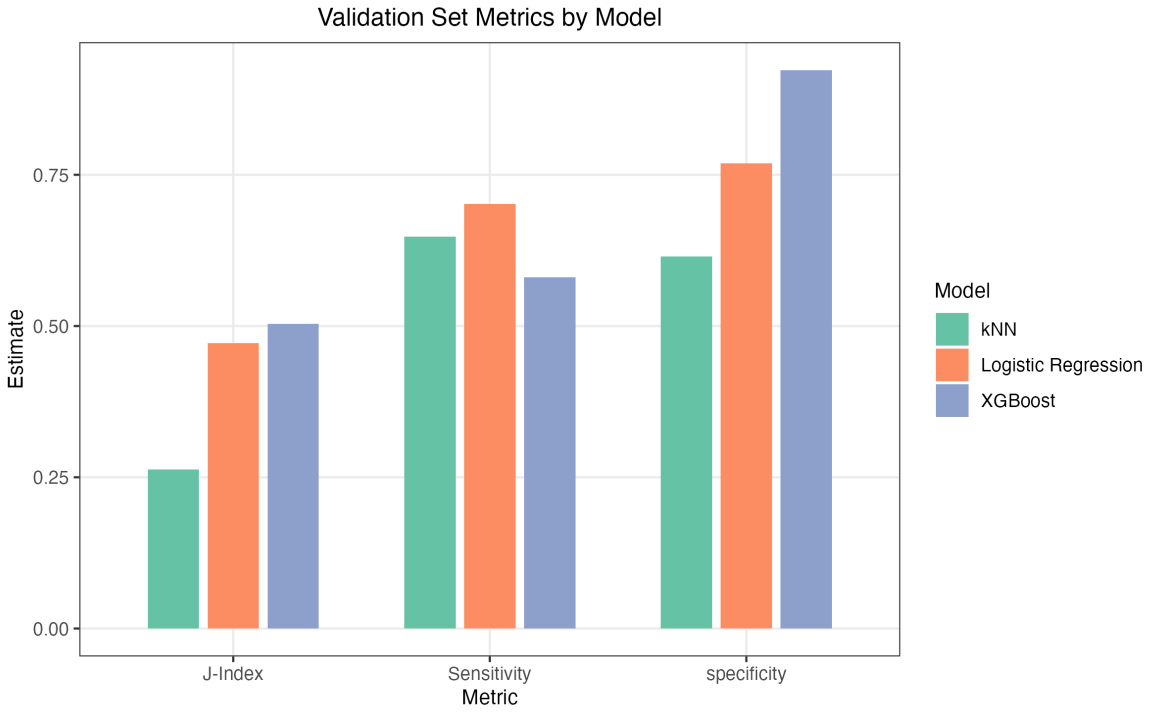

Figure 11: Validation Set Metrics by Model

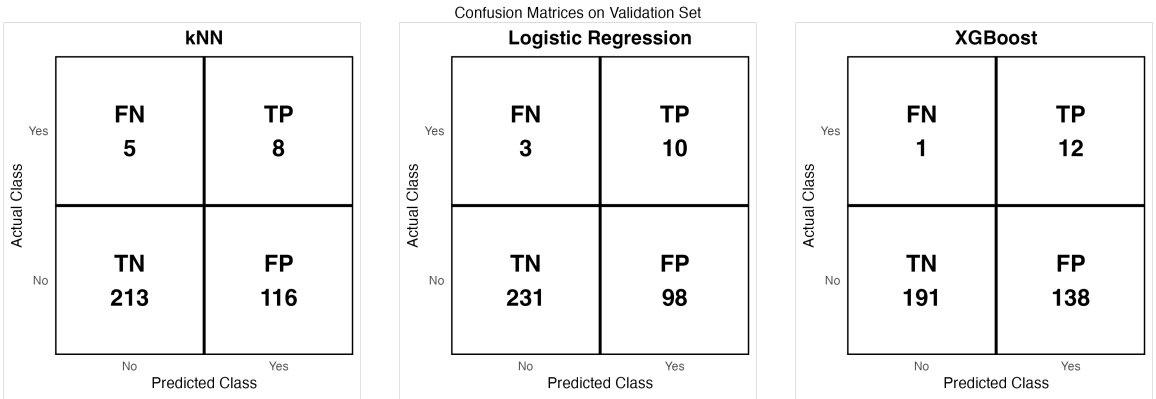

Figure 12: Confusion Matrices on Validation Set

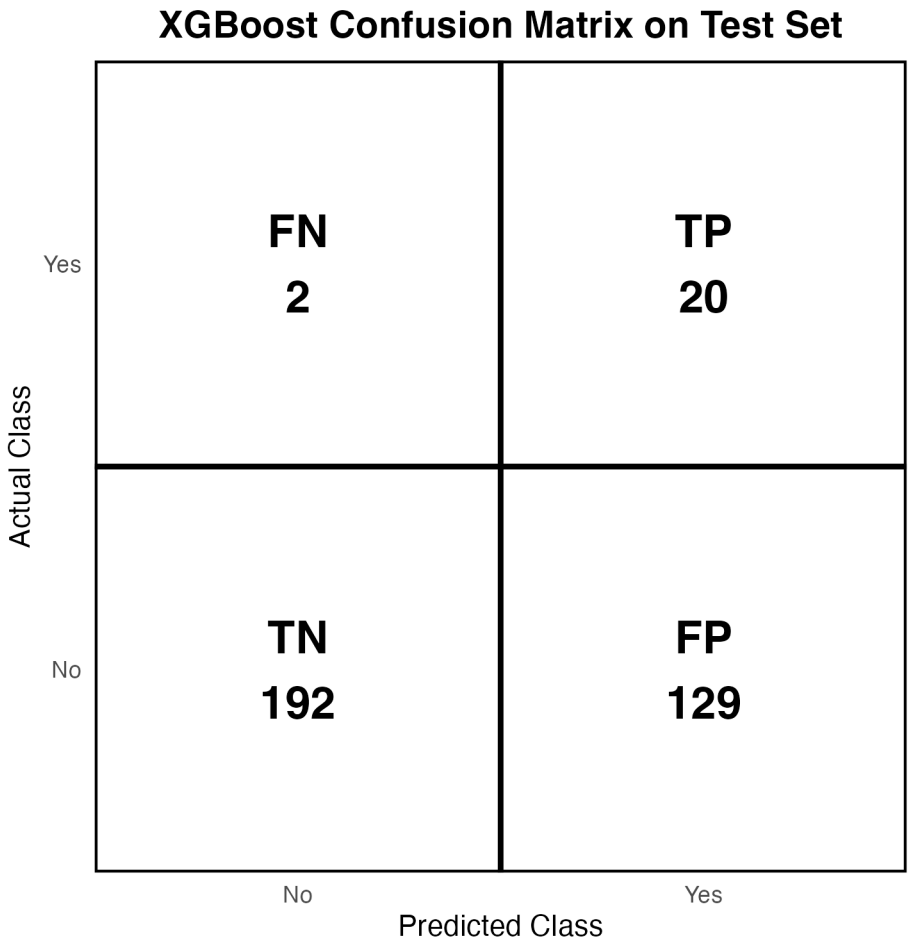

Figure 13: Final Model Confusion Matrix on Test Set

Discussion

The research project aimed to accurately predict the occurrence of stroke in patients based on a set of clinical features. The researchers utilized the Stroke Prediction Dataset, which included information on 12 variables.

Through backwards selection, age, average glucose level, and the presence of pre-existing heart disease emerged as the strongest predictors of stroke for our logistic regression model. Age was significant in determining stroke risk, with the likelihood of experiencing a stroke doubling every decade after the age of 55. Heart disease and average glucose levels were also identified as significant risk factors for stroke.

However, the research did not yield the expected results. The occurrence of false positives are quite high, and the model also predicted some false negatives. In the context of stroke risk, false positives are less serious than false negatives, while the former would still result in unnecessary usage of medical resources and costs, the latter could potentially have serious or lethal consequences.

Potential Limitations

Our current classification model focuses on determining whether a person has had a stroke based on the predictors utilized in our analysis. One limitation is that the dataset used represents a snapshot of stroke cases at a specific point in time. Additionally, our model does not account for potential risk factors that were not included in the dataset but may contribute to stroke occurrence.

Impacts

Early Intervention

By being able to predict stroke risks, healthcare providers can better identify individuals who are at a higher risk of stroke before the deadly event happens. Early identification has the potential to significantly reduce the occurrence of strokes and further complications by allowing individuals to develop personalized treatment plans with medical professionals to improve their quality of life.

Education and Awareness

Our findings can be used to educate future generations of the risk factors that are associated with having a stroke. This can encourage them to live a healthier and more positive lifestyle. As well, this may offer those who have risk factors that are genetic and not lifestyle related, a

better understanding of their risks. This will also benefit policy making. For example, if the model identifies smoking as a strong predictor of a stroke, people in public health can prioritize initiatives that better control smoking and be more specfic in their messages regarding the long term effects.

What future questions could this lead to?

One key area for future exploration is the generalizability of the model in real-word scenarios. This is specifically important in the event of insurance companies or governments deciding to use this technology in order to push their specific agendas. This may lead to unfair outcomes that discriminate or present biases based on the demographic of the original sample it was trained with. As refined and well-trained the model can be, it is crucial to consider the potential exceptions and variations that arise due to the oddities of human biology and environmental factors. Therefore this research can lead to future studies that focus on assessing the models ability across various populations. Other areas that can be explored are specific aspects of the predictors explored in this analysis. For example, one of our predictors is BMI, which is shown to be linked to all-cause mortality. However, BMI does not differentiate between fat mass and muscle mass (Cheung et al. 2021). Considering muscle mass is inversely associated with mortality (Lee and Giovannucci 2018), researchers may want to see to what degree does fat mass associate with all-cause mortality, and in our specific case, stroke.

References

Cheung, C., G. K. Lee, P. C. Au, G. H. Li, M. Chan, H. Li, B. M. Cheung, et al. 2021. "Systematic Review and Meta-Analysis of Lean Mass and Mortality: Rationale and Study Description." *Osteoporosis and Sarcopenia* 7: S3–12. https://doi.org/10.1016/j.afos.2021.01.001.

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Lee, D. H., and E. L. Giovannucci. 2018. "Body Composition and Mortality in the General Population: A Review of Epidemiologic Studies." *Experimental Biology and Medicine* 243 (17-18): 1275–85. https://doi.org/10.1177/1535370218818161.

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